

Problem Statement

Evaluate the following limit:

$$\lim_{k \rightarrow \infty} \sum_{n=2}^k \frac{(-1)^n \zeta(n)}{n}$$

Detailed Solution

This limit can be elegantly evaluated by expressing the Riemann zeta function as an infinite series and carefully exchanging the order of summation. Let's break this down step-by-step.

Step 1: Expand the Zeta Function

The limit as $k \rightarrow \infty$ represents the infinite series:

$$\sum_{n=2}^{\infty} \frac{(-1)^n \zeta(n)}{n}$$

Recall the definition of the Riemann zeta function for $n \geq 2$:

$$\zeta(n) = \sum_{m=1}^{\infty} \frac{1}{m^n} = 1 + \sum_{m=2}^{\infty} \frac{1}{m^n}$$

We isolate the $m = 1$ term because it allows us to establish absolute convergence for the remaining double sum, which is a requirement for changing the order of summation later. Substituting this expansion into our series gives:

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n} \left(1 + \sum_{m=2}^{\infty} \frac{1}{m^n} \right) = \sum_{n=2}^{\infty} \frac{(-1)^n}{n} + \sum_{n=2}^{\infty} \sum_{m=2}^{\infty} \frac{(-1)^n}{nm^n}$$

Step 2: Evaluate the First Sum

The first part of our expression is a variation of the alternating harmonic series:

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n}$$

Recall the Maclaurin series for $\ln(1+x)$:

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$$

Setting $x = 1$ yields the well-known identity for $\ln(2)$:

$$\ln(2) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

Multiplying both sides by -1 , we get:

$$-\ln(2) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -1 + \sum_{n=2}^{\infty} \frac{(-1)^n}{n}$$

Isolating our sum starting at $n = 2$, we find:

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{n} = 1 - \ln(2)$$

Step 3: Swap the Order of the Double Sum

Now we focus on the double sum:

$$\sum_{n=2}^{\infty} \sum_{m=2}^{\infty} \frac{(-1)^n}{nm^n}$$

Because the sum of the absolute values, $\sum_{n=2}^{\infty} \sum_{m=2}^{\infty} \left| \frac{(-1)^n}{nm^n} \right| = \sum_{n=2}^{\infty} \frac{\zeta(n)-1}{n}$, is convergent, we are justified by Fubini's theorem to swap the order of summation:

$$\sum_{m=2}^{\infty} \sum_{n=2}^{\infty} \frac{(-1)^n}{nm^n} = \sum_{m=2}^{\infty} \sum_{n=2}^{\infty} \frac{1}{n} \left(-\frac{1}{m} \right)^n$$

Let's evaluate the inner sum over n . Using the Maclaurin series $-\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n}$ for $|x| < 1$, we can write:

$$\sum_{n=2}^{\infty} \frac{x^n}{n} = -\ln(1-x) - x$$

Substituting $x = -\frac{1}{m}$ (which satisfies $|x| < 1$ since $m \geq 2$), the inner sum becomes:

$$\begin{aligned} \sum_{n=2}^{\infty} \frac{1}{n} \left(-\frac{1}{m} \right)^n &= -\ln \left(1 - \left(-\frac{1}{m} \right) \right) - \left(-\frac{1}{m} \right) \\ &= -\ln \left(1 + \frac{1}{m} \right) + \frac{1}{m} \\ &= \frac{1}{m} - \ln \left(\frac{m+1}{m} \right) \end{aligned}$$

Step 4: Evaluate the Telescoping Series

Substituting the evaluated inner sum back into the outer sum over m , we have:

$$\sum_{m=2}^{\infty} \left[\frac{1}{m} - (\ln(m+1) - \ln(m)) \right]$$

Let S_M be the M -th partial sum:

$$S_M = \sum_{m=2}^M \frac{1}{m} - \sum_{m=2}^M (\ln(m+1) - \ln(m))$$

The first part relates to the M -th harmonic number, $H_M = \sum_{m=1}^M \frac{1}{m}$. The second part is a telescoping sum where all intermediate natural logarithm terms cancel out, leaving only $\ln(M+1) - \ln(2)$.

$$S_M = (H_M - 1) - (\ln(M+1) - \ln(2))$$

We can cleverly group these terms by adding and subtracting $\ln(M)$:

$$S_M = (H_M - \ln(M)) - (\ln(M+1) - \ln(M)) - 1 + \ln(2)$$

$$S_M = (H_M - \ln(M)) - \ln\left(1 + \frac{1}{M}\right) - 1 + \ln(2)$$

Now, we take the limit as $M \rightarrow \infty$:

- $\lim_{M \rightarrow \infty} (H_M - \ln(M)) = \gamma$ (the Euler-Mascheroni constant)
- $\lim_{M \rightarrow \infty} \ln\left(1 + \frac{1}{M}\right) = \ln(1) = 0$

Thus, the limit of the partial sum is:

$$\lim_{M \rightarrow \infty} S_M = \gamma - 0 - 1 + \ln(2) = \gamma - 1 + \ln(2)$$

Step 5: Compute the Final Limit

Finally, we combine the separated parts from Step 2 and Step 4 to find the complete sum:

$$\begin{aligned} L &= (1 - \ln(2)) + (\gamma - 1 + \ln(2)) \\ &= 1 - \ln(2) + \gamma - 1 + \ln(2) \\ &= \gamma \end{aligned}$$

Final Answer:

$$\gamma$$